

Requirements Engineering for Trustworthy Human-AI Synergy in Software Engineering 2.0

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ABSTRACT

Software Engineering 2.0 envisions trustworthy and synergistic collaborations between humans and AI agents that are diverse, responsible, and autonomous, aiming to build the software of tomorrow – a vision that has garnered significant attention recently. Despite this growing interest, we are only beginning to unravel the complexities of fostering this synergy to develop trusted software that benefits society. This keynote will (i) highlight existing efforts to engineer the requirements for this new paradigm of software development, and (ii) chart the road ahead where Requirements Engineering can play a crucial role in defining the *sine qua nons* – the indispensable elements that ensure Software Engineering 2.0 can meet the diverse needs of various stakeholders.

CONTEXT AND KEY POINTS

Over the past two decades, the research area of AI for Software Engineering (AI4SE) has gradually emerged as a pivotal force, championing the use of specialized and customized AI to help software practitioners' daily tasks, spanning requirement analysis [1]–[3], coding [4], [5], debugging [6], [7], repair [8], [9], and beyond. In recent years, the pace of progress has accelerated significantly, particularly with the advent of deep learning methodologies [10], [11] and the emergence of large language and other foundation models [12], [13]. Recently, AI4SE has become one of the most popular research areas in the software engineering field, with dedicated conferences like the ACM SIGSOFT AI Foundational Models and Software Engineering (FORGE)¹ started in 2024. There are also many commercial offerings of AI4SE tools, e.g., GitHub CoPilot and Amazon CodeWhisperer.

A forward-looking perspective, termed Software Engineering 2.0, envisions a “*symbiotic partnership between software practitioners and autonomous, responsible, and intelligent AI4SE agents, creating a human-AI hybrid workforce*” [14]. While initial strides have been made towards this vision, c.f., [15], significant challenges persist, c.f., [16]. One critical challenge revolves around the notion that automation alone does not invariably enhance software practitioners' productivity. Research indicates that automation devoid of contextual relevance can, in fact, prevent productivity gain or even impede

productivity, c.f., [17]–[19]. Addressing this challenge requires a nuanced understanding of the contextual intricacies of AI4SE solutions when deployed alongside human practitioners.

Looking back, multiple studies exploring practitioners' perspectives have underscored such intricacies, often highlighting the divide between research outcomes and real-world needs. For instance, investigations by Lo et al. [20] and Kochhar et al. [21] have shed light on practitioners' perceptions and expectations regarding software engineering tools, including AI4SE ones, illuminating crucial insights for tool adoption. These pioneering studies have been replicated or extended to uncover the perceptions and expectations of practitioners on various research studies and tools in many areas of software engineering, including requirement engineering [22]–[26]. While some of the findings from prior studies can translate to trustworthy human-AI synergy in Software Engineering 2.0, more dedicated studies are needed.

Upon further reflection, the aforementioned studies are effectively requirement engineering endeavors, seeking to discern the prerequisites for research with practical impact on and adoption by practitioners. The Requirements Engineering (RE) area has much to offer in state-of-the-art requirements elicitation, specification, and analysis methodologies, among others. RE holds promise for shaping the next generation of AI4SE solutions, fostering trustworthy and synergistic collaborations between humans and AI towards the realization of Software Engineering 2.0. Aligned with the overarching theme of RE 2024, “Exploring New Horizons: Expanding the Frontiers of Requirements Engineering,” this keynote is a rallying cry for RE researchers. It calls for a concerted effort to uncover the indispensable requirements underpinning Software Engineering 2.0, addressing the diverse needs of stakeholders, including software practitioners, companies, users, and regulators.

BIOGRAPHY

David Lo is the OUB Chair Professor of Computer Science and Director of the Center for Research in Intelligent Software Engineering (RISE) at Singapore Management University. Championing the area of AI for Software Engineering (AI4SE) since the mid-2000s, he has demonstrated how AI — encompassing data mining, machine learning, information retrieval, natural language processing, and search-based algorithms — can transform software engineering data

¹<https://conf.researchr.org/home/forge-2024>

into actionable insights and automation. Through surveys and interviews, he has identified practitioners' pain points and explored the acceptance thresholds for AI-powered tools, effectively performing requirements engineering for AI4SE research. His contributions have led to over 20 awards, including two Test-of-Time awards and ten ACM/IEEE Distinguished Paper awards, gathering more than 30,000 citations. An ACM Fellow, IEEE Fellow, ASE Fellow, and National Research Foundation Investigator (Senior Fellow), Lo has also served as a PC Co-Chair for ASE'20, FSE'24, and ICSE'25. For more information, visit: <http://www.mysmu.edu/faculty/davidlo/>

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